

DL for ENG

LECTURE 9.2

Turbulent Flow

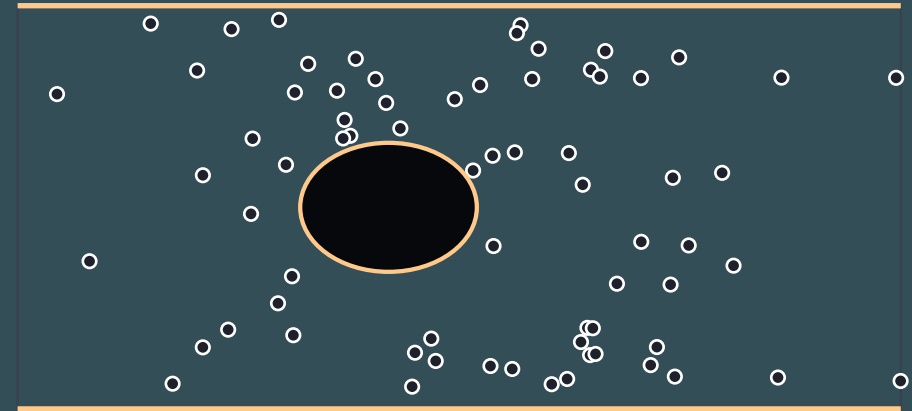
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

AVERAGE · CLOSE · ASSIMILATE



Ex_09.2: a tube in a duct, and the network's points

Above the ceiling of the last lecture the equations are averaged, and the average is not closed: a term is left that no equation determines. A model supplies it, and then the turbulent problem is the laminar one again. That is the whole trick, and the whole weakness.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Turbulence** — the range of scales, and averaging the equations
- 2 **Closure** — the term averaging leaves, and how it is modelled
- 3 **Flow with a model** — solving it, checking it, and assimilating data

Liu has no flow chapter, so the physics is Pope's and the practice is Ex_09.2's channel, where the closure is a number you choose.

READ, AND WHAT EACH ANSWERS

Pope ch. 4, 6	why can a turbulent flow not be resolved?
Pope ch. 10	what does an eddy viscosity assume?
Pope ch. 7	why is the wall the expensive part?

The Range of Scales

Turbulence — a flow that is chaotic and carries energy across a range of eddy sizes, from the width of the pipe down to the scale at which viscosity turns motion into heat. The ratio of the two grows with the Reynolds number, and so does the number of points a solver needs.

THE SMALLEST EDDY, AND WHAT IT COSTS

$$\frac{\eta}{L} \sim Re^{-3/4}$$

$$N \sim Re^{9/4} \text{ per dimension, } Re^3 \text{ in 3D}$$

η the smallest eddy, L the largest, N the grid points a solver needs

FROM THE LARGEST EDDY TO THE SMALLEST



the pipe's width

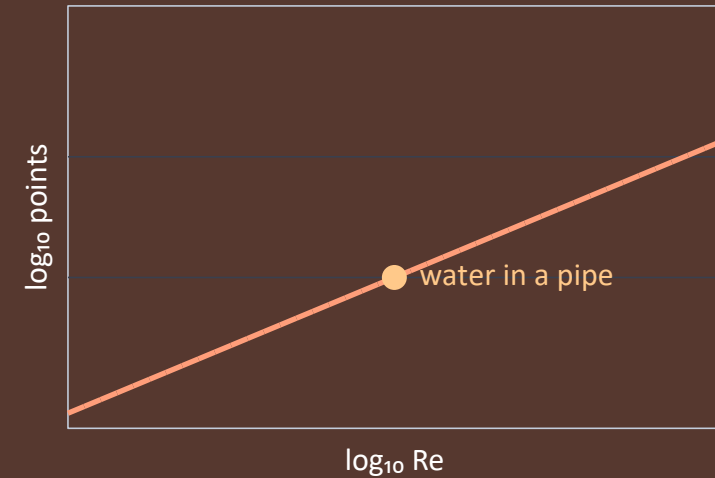
the smallest

energy enters at the largest eddies

and is handed down, size by size

the smallest turn it into heat

POINTS TO RESOLVE A 3-D FLOW



At $Re = 10^4$ that is about a billion points, and a collocation set is thousands. This is not a limit of networks: it is why turbulence modelling exists.

The Navier–Stokes Problem

Navier–Stokes problem — given a smooth initial velocity field of finite energy, do the three-dimensional incompressible equations keep a smooth solution for all time, or can the speed become unbounded in finite time? One of the seven Clay Millennium Prize problems, stated so that a smooth applied force is allowed.

THE EQUATIONS THE QUESTION IS ABOUT

$$\mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

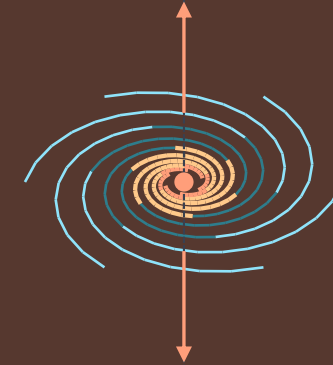
$$\nabla \cdot \mathbf{u} = 0$$

$\mathbf{u}$ velocity, p pressure, ν viscosity, $\mathbf{f}$ the applied force

WHAT WAS CLOSED, AND WHAT WAS NOT

- **closed** — a breakdown with a force, machine-checked
- **open** — the unforced question, the one people mean
- **the prize** — unclaimed, and Clay still lists it open

THE FLOW, AND WHO PROVED WHAT



a vortex spiralling inward, stretched along its axis: warm where it turns fastest, and the speed is unbounded at the core

who	what they proved
Martínez-Zorúa, Córdoba	the cascade, by hand
Alpöge, Buckmaster	forced Euler, announced first
<u>OpenAI's system</u>	unforced Euler, then forced N–S
Clay Institute	has not accepted it

Reynolds Averaging

Reynolds averaging — splitting every field into a mean and a fluctuation, then averaging the equations. What comes out looks like the flow equations for the mean, with one extra term: the average of the fluctuations multiplied together, which no equation determines.

SPLIT THE FIELD, AVERAGE THE EQUATIONS

$$u_i = \bar{u}_i + u'_i, \quad \overline{u'_i} = 0$$

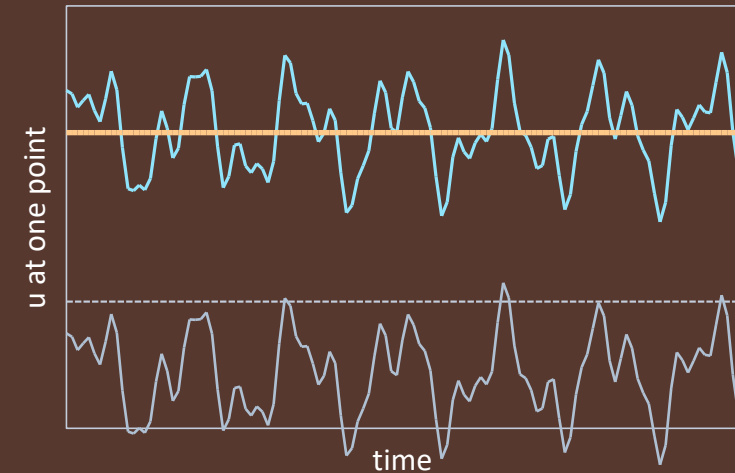
$$\bar{u}_j \partial_j \bar{u}_i = -\frac{1}{\rho} \partial_i \bar{p} + \nu \nabla^2 \bar{u}_i - \partial_j \overline{u'_i u'_j}$$

$\bar{u}_i$ the mean, u'_i the fluctuation, $\overline{u'_i u'_j}$ the Reynolds stresses, not known

THE MEAN FLOW IN PYTORCH

```
psi, p = split(net(xy))           # the MEAN field, not the flow
u, v = grad(psi, y), -grad(psi, x) # mean velocity from psi
F = conv(u, v) + grad(p, x)/rho - visc(u) + stress_term # the last one?
```

A SIGNAL, ITS MEAN, ITS FLUCTUATION



Cyan the signal at one point, amber its mean, grey the fluctuation below. The mean is what a network represents; the fluctuation carries the missing momentum.



The Closure Problem

Closure problem — the averaged equations contain the Reynolds stresses, the averaged products of the fluctuations, and there are more unknowns than equations. Writing an equation for them produces new unknown averages, so at some point a model must be supplied from outside.

MORE UNKNOWN THAN EQUATIONS

$$\tau_{ij} = -\rho \overline{u'_i u'_j}$$

τ_{ij} the Reynolds stress: six unknowns, and no equation that closes them

- **not a numerical problem** — it is in the equations
- **every solver** — CFD or PINN — needs the same model

TALK • 2 MIN How would you explain, in your own words, why averaging an equation can lose information? Then, in the card on the right, which row would you trust least on a flow with separation, and what would you do about it?

WHAT CLOSES IT, AND AT WHAT PRICE

model	supplies	costs
constant ν_t	one number	crudest
mixing length	a profile	tuned per case
two-equation	k and ϵ	two more PDEs
learned ν_t	a network	needs data

Ex_09.2 uses the first: one number for the whole duct.

THE MISSING TERM

```
F = conv(u, v) + grad(p, x)/rho - nu*lap(u)
F = F + stress_term          # unknown: supply a model
```

Eddy Viscosity

Eddy viscosity — the assumption that the Reynolds stresses act like a viscosity: proportional to the mean strain rate, through a coefficient ν_t . The averaged equations then become the laminar equations with ν replaced by $\nu + \nu_t$, and the solver of the last lecture applies again.

A MODEL THAT LOOKS LIKE VISCOSITY

$$-\overline{u'_i u'_j} = \nu_t (\partial_j \bar{u}_i + \partial_i \bar{u}_j)$$

$$\nu_{\text{eff}} = \nu + \nu_t$$

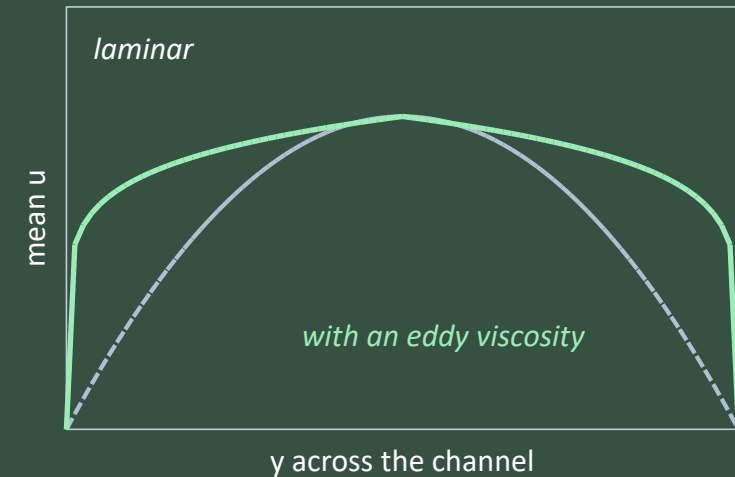
ν_t the eddy viscosity, a property of the flow; ν the fluid's own

- **what it assumes** — stresses aligned with the strain
- **where it fails** — separation, swirl, strong curvature

THE CLOSURE IN PYTORCH

```
nu_eff = nu + nu_t                # one number for the channel
F = conv(u, v) + grad(p,x)/rho - nu_eff * lap(u)  # the laminar residual
```

SCHEMATIC · THE MEAN PROFILE IT GIVES



Schematic: the model flattens the middle and steepens the wall. That steeper wall gradient is where the drag comes from, and where the points must be.

The Wall

Law of the wall — next to a wall the mean velocity follows a fixed shape when measured in wall units: y^+ for the distance and u^+ for the speed. Resolving that layer costs most of the points in any turbulent calculation, which is why a wall function replaces it with the formula.

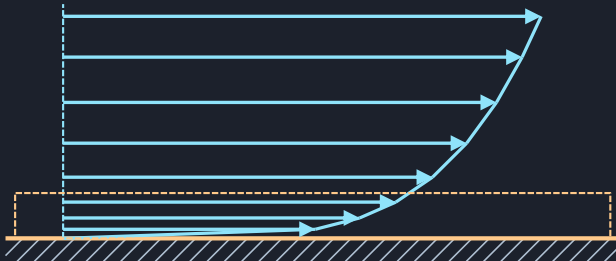
WALL UNITS, AND WHAT THEY COST

$$y^+ = \frac{u_\tau y}{\nu}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho}}$$

$$u^+ = \frac{1}{\kappa} \ln y^+ + B, \quad \kappa \approx 0.41, B \approx 5.0$$

y^+, u^+ distance and speed in wall units, u_τ the friction velocity, τ_w the wall shear

THE MEAN SPEED ABOVE A WALL

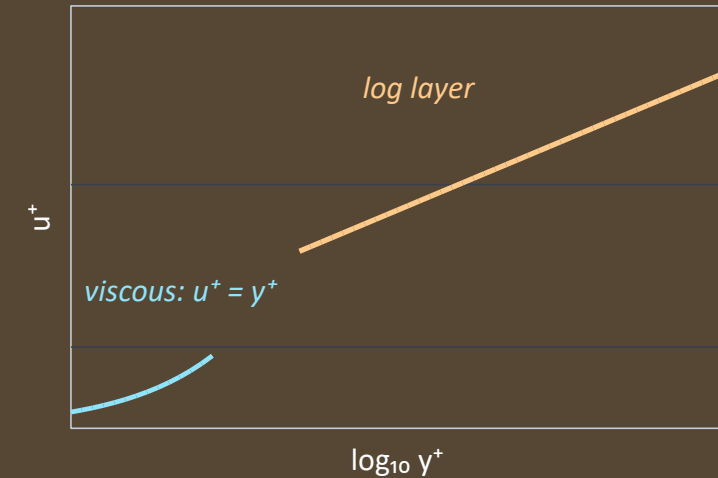


far away: hardly any change

in a thin layer it falls to zero

the points belong in that layer

THE LAW OF THE WALL



Ex_09.2 needs no grading: with its closure the model's Reynolds number is 50, the layers are thick, and half its points sit round the tube.

RANS with a PINN

RANS with a network — the averaged equations solved by the loop of L7, with the closure supplied: the network outputs the mean stream function and pressure, ν_{eff} carries the model, and the residual is the one of L9.1 with a larger viscosity.

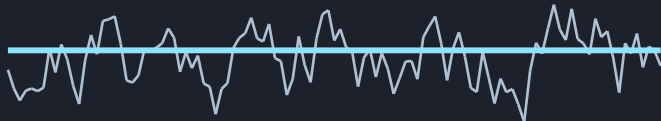
THE LAMINAR SOLVER, WITH A MODEL IN IT

$$\nu_t = \mathcal{N}_v^t(\mathbf{x}) \geq 0$$

$\mathcal{N}_v^t$ a second network, for ν_t , kept non-negative

- **unchanged from L9.1** — ψ , no-slip, the outlet, the sampling
- **changed** — one coefficient, and what it stands for

A SIGNAL, AND ITS MEAN

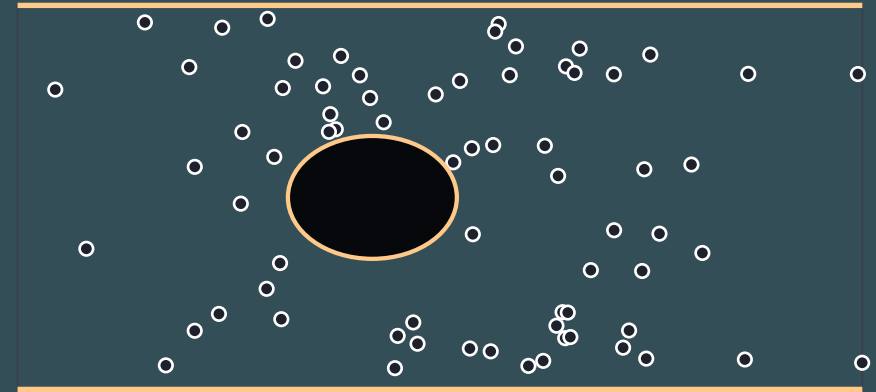


what a probe in the flow reads

its mean: what is solved for

the rest: what ν_t stands for

THE DUCT OF EX_09.2



half the points round the tube

built in	continuity, the walls, the inflow
the closure	one uniform ν_t , 500 times ν
you report	drag, and the pressure drop

Prescribe or Learn

Prescribed or learned closure — ν_t can be fixed before training, which is explicit and crude, or represented by a second network and learned, which needs data to identify it. A learned ν_t must be kept non-negative, because a negative effective viscosity makes the residual unstable.

A CHOICE, AND IT IS NOT FREE EITHER WAY

- **prescribed** — one number, stated in the report
- **learned** — a field, and it needs measurements
- **either way** — the answer inherits the model

TALK · 2 MIN How would you decide whether to fix a model coefficient or learn it, in your own words? Then, in the table on the right, which row would you defend to a client, and what would you have to show them?

THE TWO ROUTES

	prescribed	learned
ν_t is	a number you pick	a network's output
needs	a judgement	measurements
report	the number	the data and the fit
risk	wrong everywhere	right only where data is

Ex_09.2 does both: sections 4 and 5 prescribe one number, and section 6 learns that number from twelve velocity probes.

Data Assimilation

Data assimilation — fitting a flow field to sparse measurements while the averaged equations hold as a constraint. The network fills in between the sensors, recovers quantities that were never measured, such as the pressure, and needs no closure where the data speaks.

THE USE THAT A SOLVER CANNOT COPY

$$\mathcal{L} = \mathcal{L}_{RANS} + w_d \mathcal{L}_{data}$$

$\mathcal{L}_{RANS}$ the averaged equations, $\mathcal{L}_{data}$ the misfit at the sensors, w_d its weight

- **sparse velocity in** — pressure and stresses out
- **no interpolation** — the equations do the filling

BETWEEN THE SENSORS



the sensors: a few values

joined by straight lines: a guess

joined by the equations: a flow

WHAT COMES IN, AND WHAT COMES OUT



the equations constrain
the gaps

a CFD code cannot take the data as input

a PINN takes it as one more loss term

*This is the case where the method is clearly the right tool,
and it is the one to put in a proposal.*

What to Report

Reporting a turbulent solve — an engineering quantity with the model that produced it: the drag coefficient and the pressure drop, the closure and its value, the near-wall sampling, and the check that the mean flow is symmetric where it should be. A loss value is not a result.

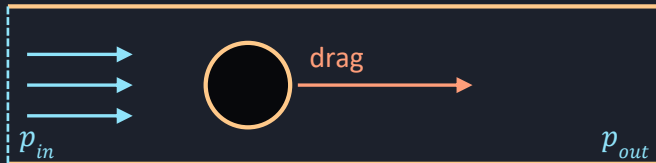
THE NUMBERS, AND WHAT MADE THEM

$$C_D = \frac{F_D}{\frac{1}{2} \rho U^2 D}, \quad \Delta p = \bar{p}_{in} - \bar{p}_{out}$$

F_D the drag force, U the mean inlet speed, D the obstacle size, $\bar{p}$ the mean pressure

- **quote with it** — the closure, and the wall sampling
- **the smooth network** — biased towards a smooth answer

WHERE THE TWO NUMBERS ARE READ



drag: the push on the tube

pressure drop: inlet minus outlet

each quoted with its closure

WHAT A NUMBER NEEDS BESIDE IT

report	without it
C_D and Δp	the result is unreadable
the closure	the result is unrepeatable
the wall points	the drag is unexplained
a symmetry check	a wrong field passes

Ex_09.2 reports both numbers three ways, against a converged reference: 218 Pa and 11.4 N per metre of tube, for one stated closure.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
two-equation models	k and ε , transported	Pope ch. 10	industrial CFD
wall functions	the log law instead of points	Pope ch. 7	high Reynolds number
large-eddy simulation	resolve the large, model the small	Pope ch. 13	unsteady loads
<u>diffusion surrogates</u>	a sampler guided by the residual	UDL ch. 14 – 18	one model, many conditions
learned closures	a model trained on data	Pope ch. 10	families of similar flows

WHY THEY ARE LEFT OUT

Ex_09.2 runs with one prescribed coefficient, which is enough to show what a closure does and what it costs. The last row is the open research question that the mini-project in L13 takes up.

Exercise 9.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — water past a tube: data, the closure, scaled
- **4** — a fitted reference, and finite volumes
- **5** — a PINN with the stream function; three compared
- **6** — the eddy viscosity, from twelve probes
- **7 – 9** — the report; mini projects MP9.2A and MP9.2B

THE QUESTION EACH PART ANSWERS

- 1 – 3** what is modelled, and how?
- 4** what is the drag, to 5 %?
- 5** how does a PINN compare?
- 6** can probes give ν_t ?

REPORT QUESTION, TO CONCLUDE

Your pressure drop and drag came from a closure that was chosen. For which task would you use the finite-element solver, and for which the network, and what would each need from you before you trusted its drag?

Key Takeaways

TURBULENCE

what it is	a range of scales, growing with Re
function	the mean is what can be solved for
train, tune	nothing: the closure is a choice
vs laminar	averaging leaves an unknown term
used for	every real flow in engineering

FLOW WITH A MODEL

what it is	the laminar solver with $v + v_t$
function	the mean field, drag, pressure drop
train, tune	points at the wall; v_t prescribed
vs a resolved solve	the answer inherits the model
used for	assimilating data, not out-solving CFD

Next — L10.1: the same method on a battery, where the unknowns are chemistry.

Questions

1 – 2 · TURBULENCE

1 What makes a turbulent flow hard to compute?

It is chaotic and carries energy across a range of eddy sizes, from the width of the pipe down to the scale at which viscosity turns motion into heat. That range grows with the Reynolds number, and so does the number of points a solver needs to resolve it. *Pope ch. 6, 9; Ex_09.2 sections 1 – 3*

2 What does Reynolds averaging do, and what is the closure problem?

Each field is split into a mean and a fluctuation and the equations are averaged. The mean obeys the flow equations with one extra term, the Reynolds stresses, which no equation determines. There are more unknowns than equations, so a model is supplied: usually an eddy viscosity ν_t . *Pope ch. 4, 10; Ex_09.2 section 3*

3 – 4 · FLOW WITH A MODEL

3 What does a RANS network solve for, and what does its answer inherit?

The mean flow: the laminar solver of L9.1 with ν replaced by $\nu + \nu_t$. It gives the mean field, the drag and the pressure drop. The answer is only as good as the closure, so a result is reported together with the model that produced it. *Pope ch. 10; Ex_09.2 sections 4 – 5*

4 What is data assimilation, and why is it the right use of a PINN on a turbulent flow?

Fitting the flow to sparse measurements while the averaged equations hold as a constraint. The network fills in between the sensors and recovers what was never measured, such as the pressure. A CFD code solves the forward problem better; it cannot take in data this way. *Ex_09.2 section 6*

NEXT

L10.1 — Batteries and Heat

The same loop again, where the unknowns are a material's chemistry and the measurements are a voltage curve.

BEFORE THE NEXT SESSION

read	Pope ch. 4, 7, 10
run	Ex_09.2, sections 1 to 7
bring	your drag, and the viscosity you found
think about	which results a client could use

Block nine is finished: a flow that can be resolved, and one that cannot.